discover compositional structure. Recognizing this pattern could inform more robust training protocols that tolerate or even encourage exploration phases, particularly in settings where emergent capabilities (reasoning, multi-step inference, compositionality) require crossing representational barriers.

Future work should investigate whether similar exploration-before-consolidation dynamics precede higher-level emergent abilities in larger models. If sentence-level or paragraph-level coherence also emerges through temporary destabilization and vocabulary churn, then Poisson-based diagnostics could serve as early-warning indicators of impending capability jumps. Conversely, understanding the conditions under which models successfully traverse barriers versus becoming trapped in suboptimal basins could guide interventions—learning rate schedules, regularization adjustments, curriculum design—that facilitate rather than suppress exploration. Our small-model study provides proof of principle that such dynamics are detectable and interpretable through statistical-mechanical frameworks, opening a path toward more principled management of the nonlinear, non-monotonic nature of neural learning.

D. OBSERVABILITY IN SMALL MODELS AND PARALLELS WITH PHYSICAL PHASE TRANSITIONS

In physical systems, phase transitions occur when order parameters change discontinuously as control variables such as temperature or pressure cross a critical threshold. Analogously, in our model, the control variable is training epoch, while linguistic statistics—dispersion, KL divergence, vocabulary count, and word length—serve as order parameters. All exhibit coordinated discontinuities in the same narrow epoch range (steps 230–250), consistent with a critical reorganization of the generative regime.

The use of multiple, independent order parameters is crucial. In physics, genuine phase transitions are characterized by the simultaneous reorganization of multiple observables—not just one. Here, dispersion and KL divergence measure fundamentally different aspects of the count distributions (variance structure vs. full distributional form), yet both display synchronized cusps. This convergence across non-redundant metrics mirrors how physical phase transitions are confirmed through multiple experimental probes (e.g., heat capacity, magnetization, correlation length), each sensitive to different aspects of the critical reorganization.

This behavior parallels two known phenomena: grokking, where small models abruptly transition from memorization to generalization on algorithmic tasks, and emergent abilities in large language models, where new capabilities appear suddenly beyond critical scale thresholds. By observing these reorganizations in a 3.6M-parameter transformer trained on a modest corpus, we directly address the first research question: phase-transition-like behavior is not exclusive to large-scale systems or massive models observed through log-scaled compute. Such discontinuities can manifest in small, tractable models when analyzed with appropriate statistical probes. This reinforces the idea that emergent dynamics are scale-invariant in form—they represent fundamental properties of neural learning, not mere artifacts of size. Poisson-based diagnostics—combining both dispersion and KL divergence—thus provide a unifying framework for linking the nonlinear dynamics of learning to the physics of critical phenomena.

E. ADDRESSING THE SCHAEFFER ET AL. CRITIQUE

Our methodological approach directly addresses concerns raised by Schaeffer, Miranda, and Koyejo regarding the reality of emergent abilities. They demonstrated that many reported emergent capabilities arise from discrete, non-linear evaluation metrics rather than genuine phase transitions, and that continuous metrics often reveal smooth scaling instead of discontinuities. Our study heeds this critique by employing continuous, internal statistical measures—dispersion and KL divergence—rather than binary task accuracy thresholds.

Critically, we detect synchronized discontinuities across multiple independent continuous metrics, each measuring different statistical properties. Dispersion quantifies variance relative to the mean; KL divergence captures full distributional deviation; word length measures linguistic complexity; vocabulary dynamics track compositional reorganization. No single metric alone would suffice to claim a phase transition, but their simultaneous cusps within the same narrow epoch window provide converging evidence that cannot be explained by metric artifacts. Furthermore, these signatures appear in raw training space without rescaling, and are invisible in standard loss curves, confirming that we are detecting internal reorganization rather than external scoring effects.

F. LIMITATIONS AND SCOPE

While the findings provide convergent evidence for a small-model phase transition, several limitations warrant caution. First, we analyzed a single architecture (3.6M parameters) and a single dataset (Tiny Shakespeare); generalization to larger models, multilingual corpora, or instruction-tuned datasets remains untested. Second, character-level tokenization differs from subword or BPE segmentation used in practical LLMs, and the correct/incorrect heuristic may misclassify rare but valid words or neologisms. Third, our study focuses on external statistical signatures—dispersion, KL divergence, vocabulary composition—while the correspondence between these metrics and internal representational shifts (such as activation clustering, attention alignment, or hidden-state geometry) remains open. Fourth, alternative decoding methods (top-k, nucleus sampling, beam search) might influence surface statistics and potentially shift the apparent transition region, though we expect the qualitative reorganization to persist. Fifth, we have not yet examined universality across model sizes, datasets, or architectural families; testing whether the same Poisson $\rightarrow$ sub-Poisson signature recurs under variation would strengthen claims of genuine criticality. Finally, while we employ two complementary Poisson-based metrics—dispersion and KL divergence—future work should explore whether other statistical probes (e.g., higher-order cumulants, entropy measures, or correlation functions) reveal additional structure during the transition. The robustness of the synchronized cusps across dispersion and KL suggests the transition is genuine, but expanding the diagnostic toolkit would further strengthen this interpretation.

VI. FUTURE WORK

Future research should extend these analyses along four complementary directions.

(1) Scaling and universality: Test whether Poisson–sub-Poisson reorganizations recur across model scales (from small to billion-parameter systems), diverse datasets (multilingual corpora, code, instruction-tuned data), and architectural families (encoder-decoder, causal transformers, state-space models). Establish whether the transition epoch, dispersion dynamics, and KL divergence cusps exhibit universal patterns or depend systematically on scale and domain. This would assess whether our findings reflect a general property of language model training or are specific to small character-level systems.

(2) Quantitative characterization: Apply finite-size scaling analysis to distinguish genuine critical phenomena—characterized by power-law divergences and universal scaling exponents—from simple threshold effects. Estimate critical exponents for dispersion, KL divergence, and correlation lengths near the transition. Determine whether the transition is first-order (discontinuous order parameter, hysteresis, mixed phases) or continuous (critical slowing, diverging fluctuations), following the framework of Rubin, Seroussi, and Ringel. This would rigorously classify the transition within the taxonomy of statistical mechanics.

(3) Mechanistic linkage: Connect external Poisson-based statistics with internal network dynamics. Investigate whether dispersion cusps coincide with changes in hidden-state correlation lengths, attention-map reorganizations, embedding-space clustering, or gradient flow patterns. Use mechanistic interpretability techniques—such as singular value decomposition of representations, neuron activation analysis, and circuit tracing—to identify which internal structures reorganize during the transition. This would bridge behavioral signatures (dispersion, KL) with representational mechanisms, revealing how the model reorganizes internally to produce the observed statistical shift.

(4) Higher-level transitions: Investigate whether similar Poisson–sub-Poisson discontinuities appear at sentence or paragraph levels, signaling the sudden emergence of grammatical coherence, semantic consistency, or discourse structure. Extend the windowing framework to multi-word sequences and track dispersion/KL divergence for syntactic constructions (e.g., subject-verb agreement, clause embedding). Test whether linguistic hierarchies exhibit cascading transitions—lexical coherence first, then syntactic, then semantic—each marked by distinct reorganization signatures.

Together, these efforts would test whether Poisson-centered metrics provide a general signature of emergent coherence across linguistic hierarchies, model scales, and training regimes.

Taken together, these findings answer the three motivating questions posed in the introduction with converging evidence: (1) phase-transition-like reorganizations are not unique to large models but occur in small transformers when probed with appropriate statistical metrics; (2) they can be detected directly in linear training space without logarithmic rescaling, using continuous Poisson-based diagnostics rather than binary task thresholds; and (3) they emerge at early stages of training—around epochs 230–250 in our 3.6M-parameter model—as lexical coherence forms, well before loss convergence. The synchronization of multiple independent metrics—dispersion, KL divergence, word length, and vocabulary dynamics—within the same narrow epoch window provides robust evidence that these phenomena reflect genuine internal reorganizations, not metric artifacts or gradual drift. These results collectively demonstrate that emergent phenomena are intrinsic, measurable properties of neural language model training dynamics, observable even at modest scale.

VII. CONCLUSION

We have presented converging evidence that a small transformer undergoes a distinct, phase-transition-like reorganization during training, detectable directly in linear training space through Poisson-centered diagnostics. Both dispersion and KL divergence reveal synchronized cusps at epochs 230–250, where correct-word statistics briefly return toward Poissonian behavior before settling into sub-Poisson regularity. Unique incorrect vocabulary first expands during fragment proliferation, then collapses as errors are abandoned, while the unique correct vocabulary grows monotonically. Average correct-word length exhibits a sharp S-curve increase from $\sim$1.5 to $\sim$2.5 characters, and prefix tracking of the word "you" illustrates how single- and two-letter fragments consolidate into a stable three-letter form precisely at the transition epoch.

Together, these findings demonstrate that emergent-like reorganizations can arise even in modest transformer models at early stages of training, without requiring log-scaling or massive compute. By employing multiple independent, continuous statistical probes—dispersion (variance structure), KL divergence (full distributional form), word length (linguistic complexity), and vocabulary dynamics (compositional reorganization)—we establish a quantitative framework for detecting discontinuous learning dynamics that avoids the metric artifacts identified by recent critiques of emergent abilities. The synchronization of cusps across non-redundant metrics provides robust evidence for genuine internal restructuring.

This approach bridges concepts from statistical physics with linguistic emergence, showing that coherence at higher linguistic levels—words, sentences, and beyond—may likewise arise through abrupt structural reorganizations rather than gradual accumulation. Small models thus serve as powerful laboratories for studying criticality in neural systems, and Poisson-based diagnostics provide a unifying lens for understanding the nonlinear dynamics of learning in language models. Our results suggest that phase-transition-like phenomena are fundamental properties of neural training, observable at any scale when appropriate probes reveal the underlying reorganization.

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